

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week-2

Create Database

```
MySQL 8.0 Command Line Cli  x  +  v

Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE bookflow_week2;
Query OK, 1 row affected (0.02 sec)

mysql> USE bookflow_week2;
Database changed
```

Create Books Table

```
mysql> CREATE TABLE Books (
->     book_id INT PRIMARY KEY,
->     title VARCHAR(100) NOT NULL,
->     isbn VARCHAR(20) UNIQUE,
->     published_year INT CHECK (published_year < 2027)
-> );
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
-> (1, 'The Great Gatsby', '9780743273565', 1925),
-> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
-> (3, '1984', '9780451524935', 1949);
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

Books Table Output

```
mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title                | isbn          | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby     | 9780743273565 | 1925           |
| 2       | To Kill a Mockingbird | 9780061120084 | 1960           |
| 3       | 1984                 | 9780451524935 | 1949           |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Create Members Table

```
mysql> CREATE TABLE Members (
  ->     member_id INT PRIMARY KEY,
  ->     full_name VARCHAR(100),
  ->     email VARCHAR(100) UNIQUE
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO Members (member_id, full_name, email) VALUES
  -> (101, 'John Smith', 'john.smith@email.com'),
  -> (102, 'Emma Wilson', 'emma.wilson@email.com'),
  -> (103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

Members Table Output

```
mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name      | email                      |
+-----+-----+-----+
| 101       | John Smith     | john.smith@email.com       |
| 102       | Emma Wilson    | emma.wilson@email.com      |
| 103       | Michael Brown  | michael.brown@email.com    |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

Create Loans Table

```
mysql> CREATE TABLE Loans (
->     loan_id INT PRIMARY KEY,
->     member_id INT,
->     book_id INT,
->     loan_date DATE,
->     FOREIGN KEY (member_id) REFERENCES Members(member_id),
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
-> (1,101,1, '2025-01-05'),
-> (2,102,2, '2025-01-08'),
-> (3,103,3, '2025-01-10'),
-> (4,101,2, '2025-02-01'),
-> (5,102,1, '2025-02-05'),
-> (6,103,2, '2025-02-12'),
-> (7,101,3, '2025-03-01'),
-> (8,102,3, '2025-03-07'),
-> (9,103,1, '2025-03-15'),
-> (10,101,1, '2025-04-01');
Query OK, 10 rows affected (0.02 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

Loans Table Output

```
mysql> SELECT * FROM Loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 102 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
| 6 | 103 | 2 | 2025-02-12 |
| 7 | 101 | 3 | 2025-03-01 |
| 8 | 102 | 3 | 2025-03-07 |
| 9 | 103 | 1 | 2025-03-15 |
| 10 | 101 | 1 | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

INNER JOIN Output

```
mysql> SELECT
-> m.full_name AS Member_Name,
-> b.title AS Book_Title
-> FROM Loans l
-> INNER JOIN Members m
-> ON l.member_id=m.member_id
-> INNER JOIN Books b
-> ON l.book_id=b.book_id;
```

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984
Michael Brown	1984
Michael Brown	To Kill a Mockingbird
Michael Brown	The Great Gatsby

10 rows in set (0.00 sec)

GROUP BY Output

```
mysql> SELECT
-> published_year,
-> COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_Books
1925	1
1949	1
1960	1

3 rows in set (0.01 sec)

Create Donation_History Table

```
mysql> CREATE TABLE Donation_History (  
-> donation_id INT PRIMARY KEY,  
-> book_id INT,  
-> donor_name VARCHAR(100),  
-> donation_date DATE,  
-> FOREIGN KEY(book_id)  
-> REFERENCES Books(book_id)  
-> );  
Query OK, 0 rows affected (0.08 sec)
```

Transaction Output

```
mysql> START TRANSACTION;  
Query OK, 0 rows affected (0.00 sec)
```

Books + Donation_History Output

```
mysql> INSERT INTO Books
-> (book_id,title,isbn,published_year)
-> VALUES
-> (4,'Animal Farm','9780451526342',1945);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO Donation_History
-> (donation_id,book_id,donor_name,donation_date)
-> VALUES
-> (1,4,'Raj Kumar',CURDATE());
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title                | isbn          | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby     | 9780743273565 | 1925           |
| 2       | To Kill a Mockingbird | 9780061120084 | 1960           |
| 3       | 1984                 | 9780451524935 | 1949           |
| 4       | Animal Farm          | 9780451526342 | 1945           |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Donation_History;
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
| 1           | 4       | Raj Kumar  | 2026-07-30    |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Create Index

```
mysql> CREATE INDEX idx_books_isbn
-> ON Books(isbn);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Search by ISBN Output

```
mysql> SELECT *  
      -> FROM Books  
      -> WHERE isbn='9780451524935';  
+-----+-----+-----+-----+  
| book_id | title | isbn          | published_year |  
+-----+-----+-----+-----+  
|        3 | 1984  | 9780451524935 |          1949 |  
+-----+-----+-----+-----+  
1 row in set (0.00 sec)  
  
mysql>
```